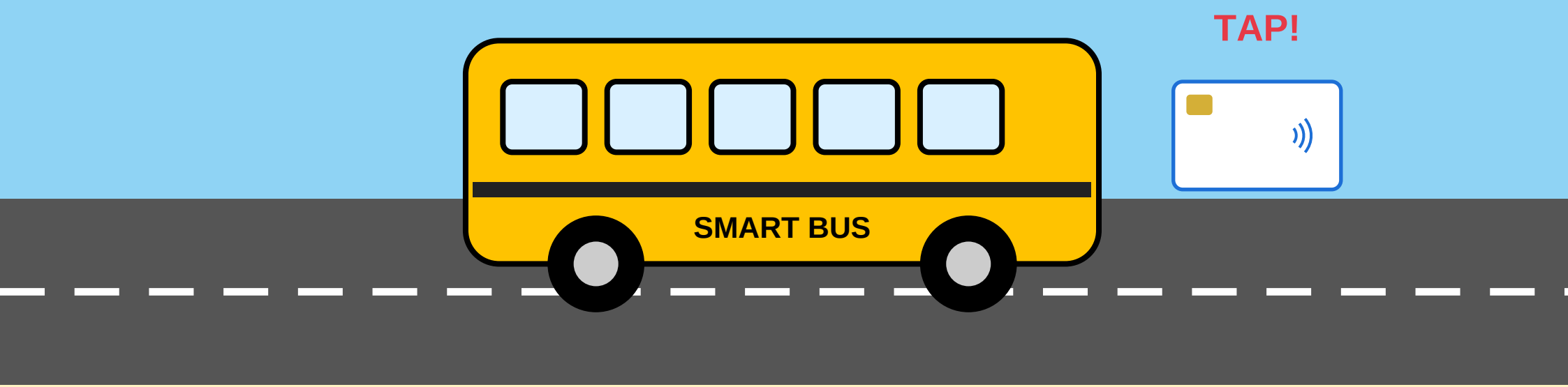
BUILD A SMART BUS TICKET MACHINE!

A picture-by-picture guide for young makers



What you will make:

Press the driver button to pick the stop. Tap an RFID card to ride.

The screen and LEDs say YES or NO. A Wi-Fi website shows who rode!

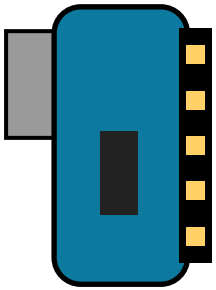
SAFETY

Ask a grown-up to help.

POWER OFF when you wire.

MEET YOUR PARTS

Every part has a job. Say hello to them before you start!

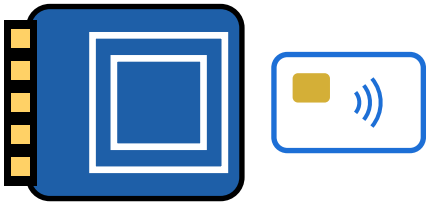


Arduino Uno

The BRAIN of the bus.
Reads cards, runs screen.

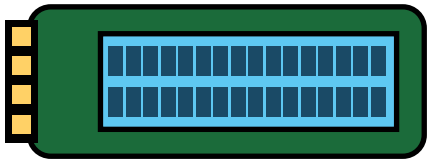
NodeMCU

The WI-FI helper.
Counts money, makes website.



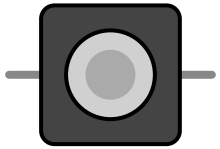
RC522 + Card

The CARD READER.
Says who you are (3.3V only!).



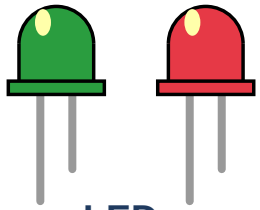
LCD Screen

The TALKING screen.
Shows stop and result.



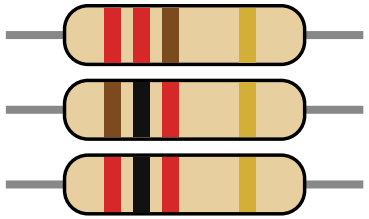
Button

The DRIVER button.
Press = next stop.



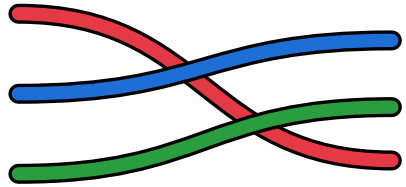
LEDs

Green = YES.
Red = NO.



Resistors

Tiny speed bumps
for electricity.



Jumper Wires

Cables that join
the parts together.

Also: a breadboard (a plug-in board for resistors), a laptop with Arduino IDE, and a USB cable for each board.

RESISTORS AND THE VOLTAGE DIVIDER

Our RC522 and NodeMCU are shy: they only like 3.3V. The Uno shouts at 5V. We make the shout smaller!

Read the colour stripes

220 ohm red, red, brown

Protects the LEDs (need 2)

1k ohm brown, black, red

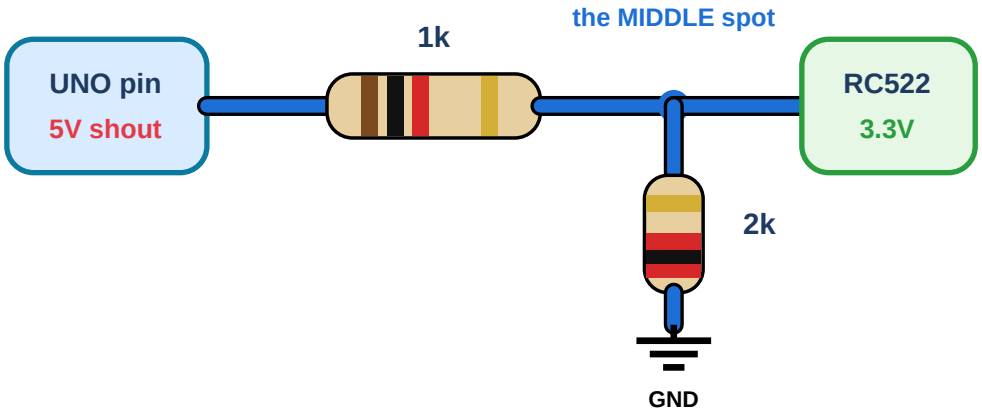
First speed bump (need 5)

2k ohm red, black, red

Second speed bump (need 5)

Gold stripe = how exact it is. Ignore it!

The voltage divider



The maths (it is easy!)

5V x 2k / (1k + 2k) = 5V x 2/3 = about 3.3V

The 2k bump keeps 2 parts out of 3. Safe!

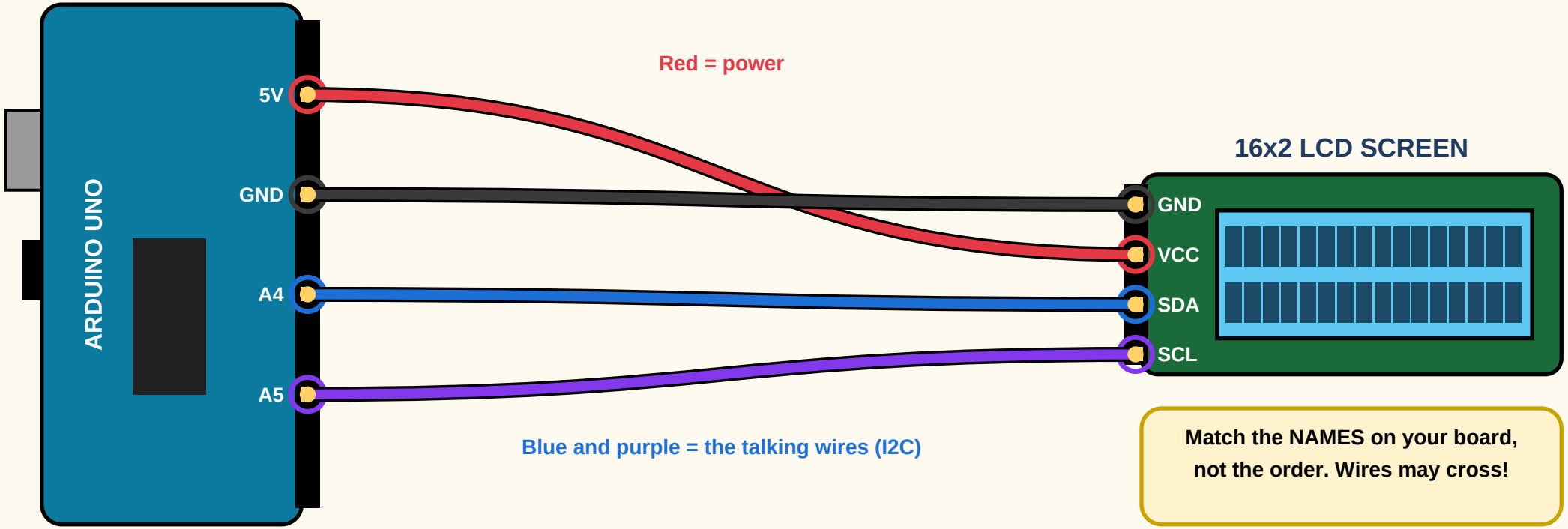
You will build 5 dividers in total.

Slide picture: the 1k slows the water, the 2k lets most of it drain to GND.

Only the Uno-to-shy-chip wires need it!

STEP 1: WIRE THE LCD SCREEN

The screen only needs 4 wires. Power on top, talking wires below.

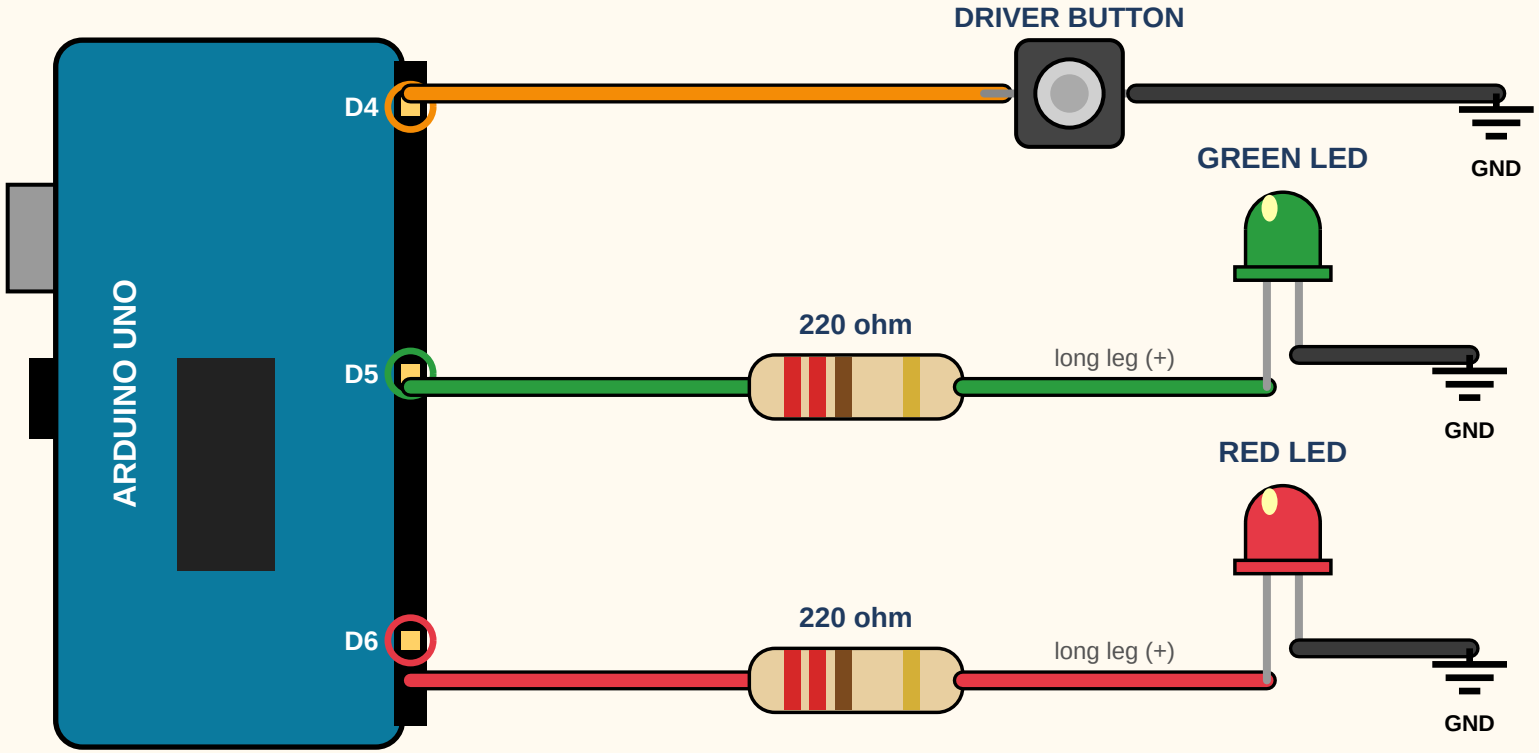


DO IT LIKE THIS

1. Unplug everything. Put the screen where you can see its 4 pins.
2. RED wire: Uno 5V to LCD VCC.
3. BLACK wire: Uno GND to LCD GND.
4. BLUE wire: Uno A4 to LCD SDA. PURPLE wire: Uno A5 to LCD SCL.
5. Later, if the screen is blank, turn the little blue screw on the back (contrast).

STEP 2: BUTTON AND LEDs

The button picks the stop. The green LED says YES and the red LED says NO.



LED TIPS

The LONG leg (+) goes toward the resistor and Uno pin. The SHORT leg (-) goes to GND.

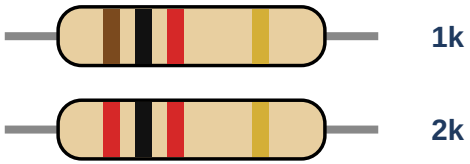
DO IT LIKE THIS

1. ORANGE wire: Uno D4 to the button. The other button leg goes to GND.
2. GREEN wire: Uno D5 to a 220 ohm resistor, then to the green LED's long leg.
3. RED wire: Uno D6 to a 220 ohm resistor, then to the red LED's long leg.
4. Short legs of both LEDs go to GND. (A GND sign means: wire it to a GND pin.)
5. Never forget the resistor! Without it the LED can burn out.

STEP 3: BUILD ONE VOLTAGE DIVIDER

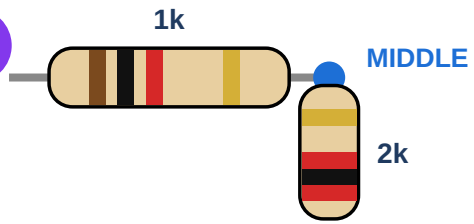
You will copy this 5 times. Practice once here, then repeat!

1



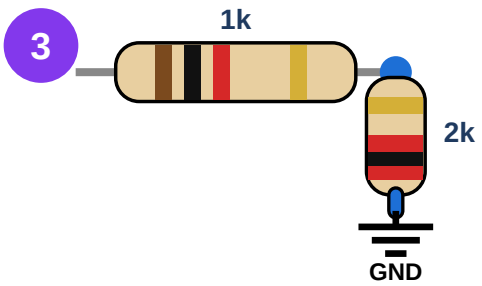
Pick one 1k and one 2k resistor.

2



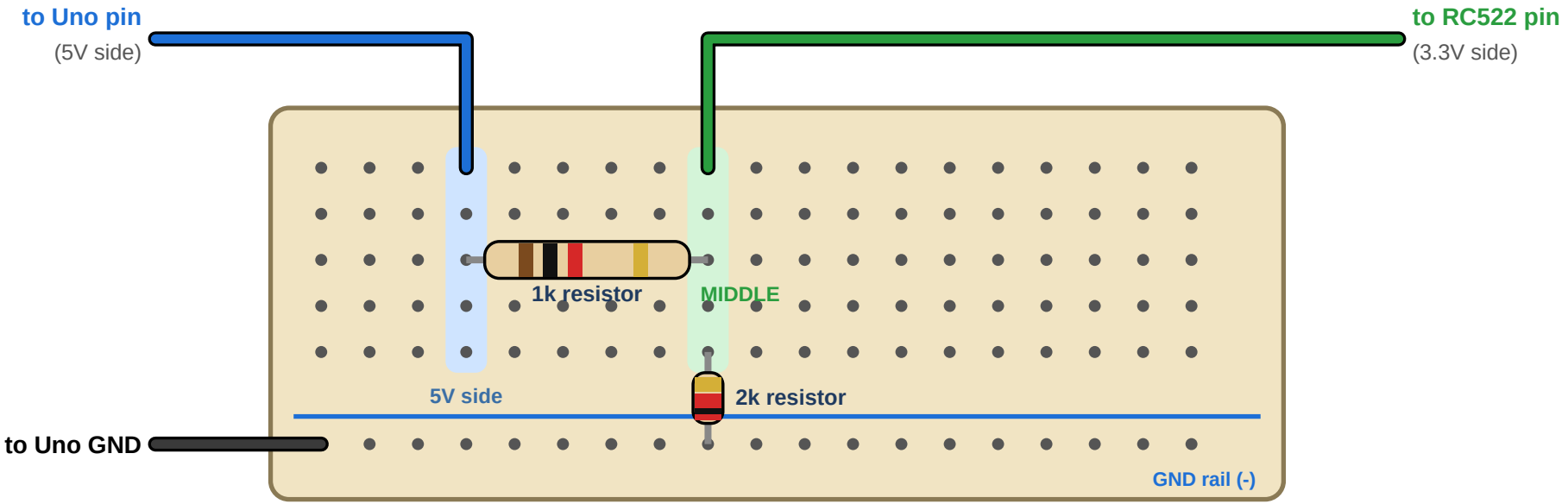
Join one leg of each.
This meeting spot is the MIDDLE.

3



Connect the free 2k leg to GND.

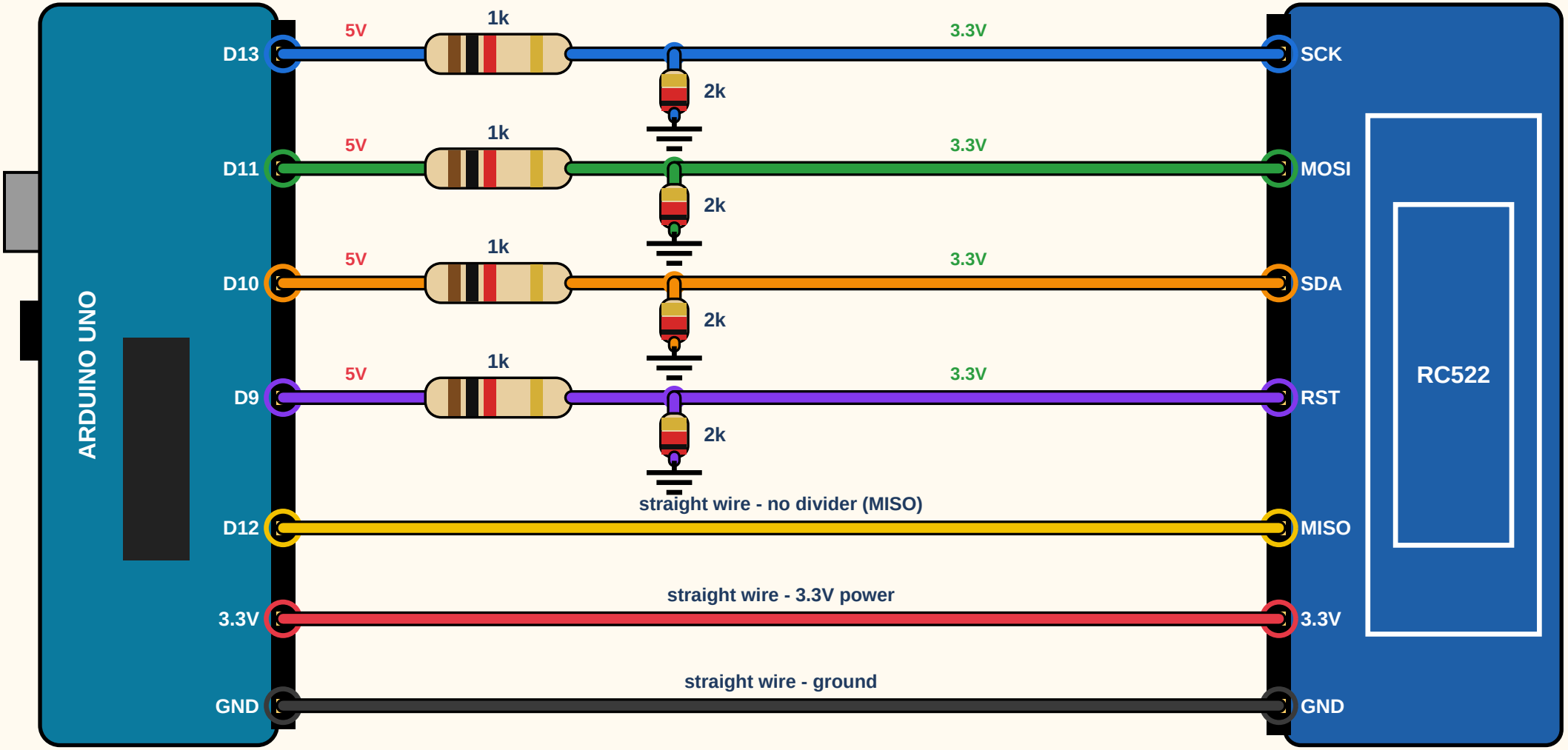
4 **Put it on the breadboard exactly like this:**



Holes in the same short column are joined inside the board. The 1k and the 2k meet in the MIDDLE column.

STEP 4: WIRE THE RFID CARD READER (RC522)

4 wires need a divider. 3 wires go straight. The RC522 is shy: 3.3V only!

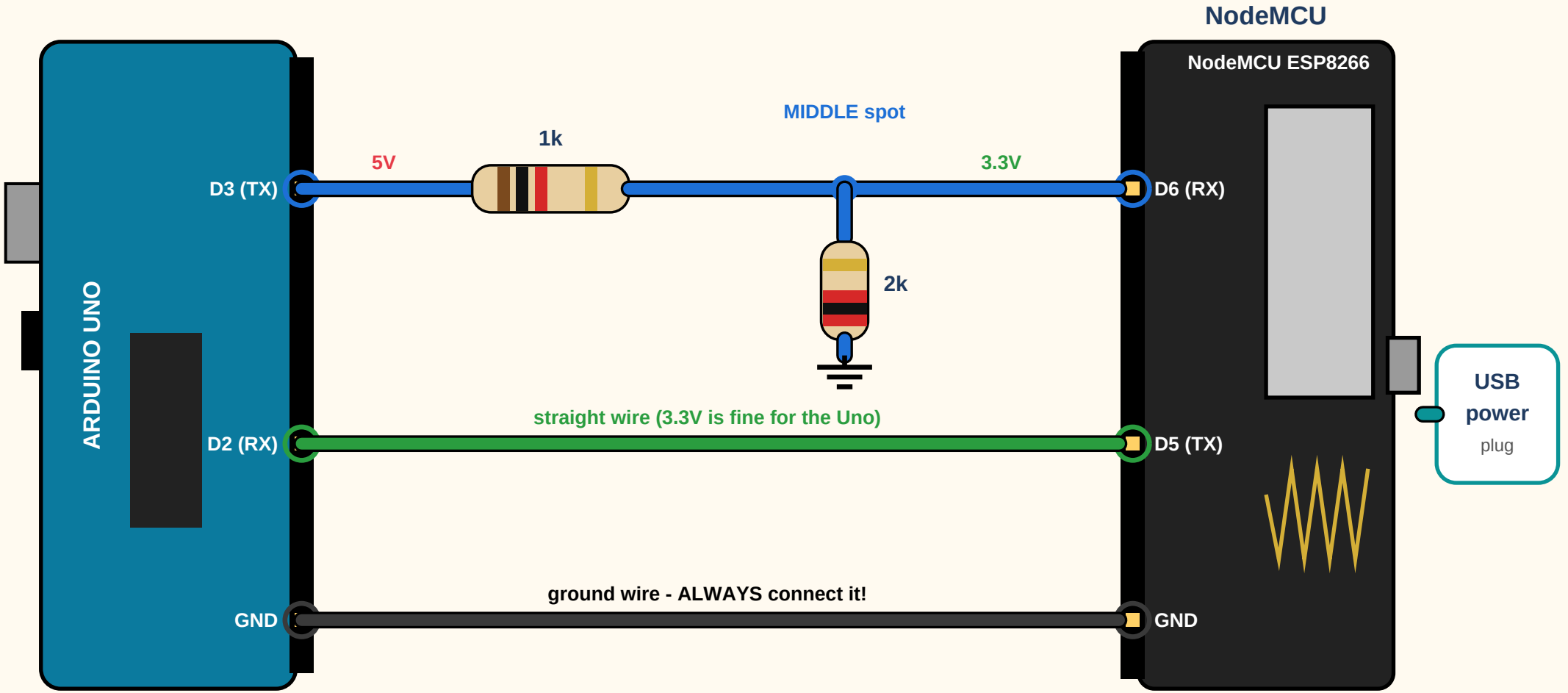


DO IT LIKE THIS

1. Build 4 dividers (page 6) for D13, D11, D10 and D9. The 2k leg of each goes to GND.
2. Plain wires, NO divider: D12 to MISO, Uno 3.3V to RC522 3.3V, GND to GND.
3. NEVER connect the RC522 to 5V! Leave the IRQ pin empty. Match pin NAMES on your board.

STEP 5: UNO TALKS TO NodeMCU

Two wires to talk, one wire to share ground. The Uno's shout needs a divider again!



DO IT LIKE THIS

1. BLUE: Uno D3 to 1k, middle spot to NodeMCU D6.
2. 2k from the middle spot to GND.
3. GREEN: NodeMCU D5 to Uno D2.
4. BLACK: Uno GND to NodeMCU GND.

CAREFUL!

NodeMCU gets its OWN USB power.
Never use the Uno 3.3V pin for it.

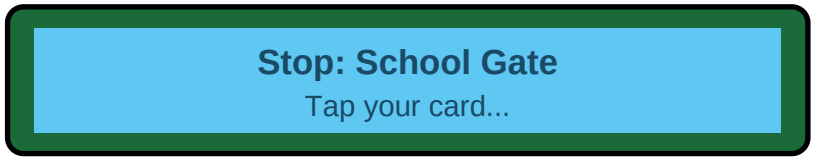
STEP 6: UPLOAD THE CODE AND TRY IT

Now bring your bus to life!

UPLOAD

1. Install the Arduino IDE.
2. Add libraries: MFRC522 and LiquidCrystal_I2C.
3. Add the ESP8266 board package.
4. In nodemcu_smart_bus.ino type your Wi-Fi name
5. and password. Put in your card UIDs. Upload.
6. Upload arduino_smart_bus.ino to the Uno.
7. Open Serial Monitor (115200) and note the IP.
8. On a phone on the same Wi-Fi, open that IP.

TEST IT



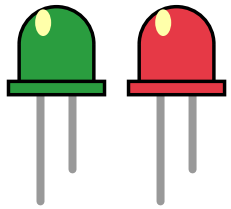
Short press = next stop.

Hold 2 seconds = confirm stop.

Tap card = board. Tap again = leave.

Green LED = approved. Red LED = declined.

Unknown card = red light.



Checklist before you celebrate

- ☐ Power and ground joined
- ☐ Card UID is read
- ☐ Boarding and exit tap work
- ☐ Website opens
- ☐ LCD shows text
- ☐ Button short + long press work
- ☐ Unknown card tested
- ☐ Power off: balances reset (normal!)

GOLDEN RULES AND DETECTIVE CORNER

Stay safe, and fix problems like a detective.

THE 6 NEVERS

- Never put 5V on the RC522 or NodeMCU pins.
- Never wire Uno D3 straight to NodeMCU RX.
- Never leave out the LED resistors.
- Never forget to join GND of both boards.
- Never move wires while the power is ON.
- Never use real bank or debit card info.

Remember: card numbers can be copied. This is a classroom toy.

DETECTIVE CORNER

Card reader not found?
Check 3.3V, the 4 dividers and GND.

LCD is blank?
Check SDA/SCL, address 0x27, contrast screw.

NodeMCU keeps restarting?
Use a steady USB power source.

Boards do not talk?
Check D2/D3, D5/D6, GND, speeds, divider.

Card says unknown?
Put its UID in the NodeMCU code.

You did it! Now ride the Smart Bus!